

DTU



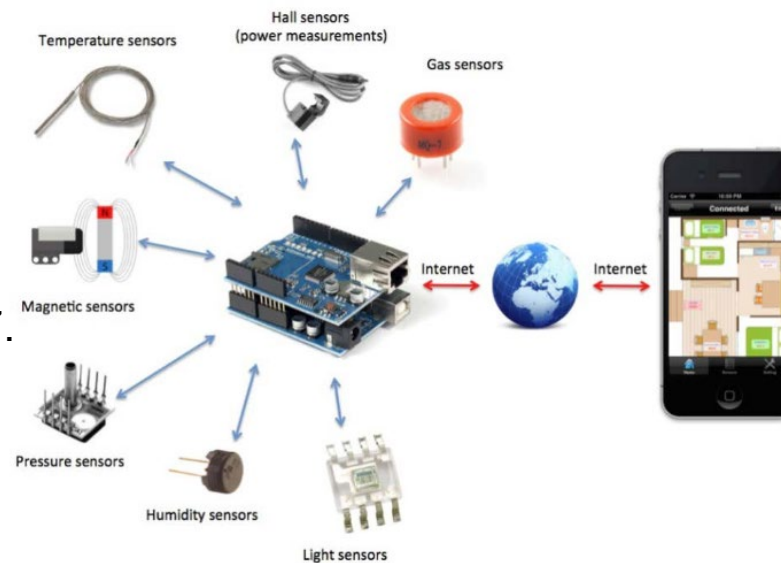
34315 – IoT application and infrastructure implementation
F26

Presentation info

Requirements

- Use of WiFi communication (i.e. ESP8266)
- Minimum 1 analog sensor, 1 digital sensor and one form of actuation
- Backend/visualization: Minimum web server or Thingspeak, and one form of display on device
- Poster pitch and demo to the class (7. May – last lecture day)
- Report: Max 10 page documentation and description plus appendices (log book, group contract, selv-evaluation, videos, code) (due 17. May)

Feel free to expand with all components in the box. Be creative and play with IoT. Think of your project as a service that could be sold to customers.



Presentation content and procedures

- **Poster presentation**
- Presentation in class on 7. May
- Max 8 minutes per group incl. a demo (consider video vs. live)
- Everybody must be active in the presentation in turn, and other groups will give feedback

Upload to DTU LEARN by 17. May

- Presentation poster incl. video
- Report incl. appendix
- All code as a zip file

Make sure to include all files – we will not contact you for adding missing content after the deadline.

Deadline is firm !!

- **Report**
- 17. May
- 10 page project brief
 - Executive summary
 - Pain (what problem are we trying to solve)
 - Solution
 - Technical highlights, risks, key challenges, system design, etc
 - How to present data to the user
- Appendix:
 - Who did what (include percentage distribution)
 - Bigger figures for readability
 - Log-book
 - Group contract
 - Self-evaluation – individual and per group

The poster session

- 7. May between 8.00 – 13.00, everybody must be present for the entire session
- Counts as the oral exam
- 8 minute poster presentation incl. Demo
- Each student must give the presentation individually
 - i.e. everybody must be able to give the presentation and execute the demo
- Other group members will ask questions and give feedback to other poster (3 min)
 - May not attend same poster as a group mate



Remember: take all necessary videos and fotos, as ESPs and other borrowed components must be handed in after the presentation on 7. May

How to approach your project - checklist

- ☐ Identify the problem.
- ☐ Formulate goals. Formulate technical specification, if possible, that can help define the success of the project. Be as specific as possible. Measurable metrics to the extent possible.
- ☐ Acquiring basic necessary knowledge and studying background theory.
- ☐ Survey to find out what other people have done. This can often amount to a state-of-the-art section in the report.
- ☐ Define several different possible solutions to the problem (based on theoretical studies and survey).
- ☐ Chose a solution to implement and consider limitations when choosing (time, cost, complexity, etc.). This includes your technology-choices, for which you need solid arguments, preferably backed up by references.
- ☐ Translate original soft goal into “hard” & specific, and measurable requirements
- ☐ Now you can design your system (block diagram, interfaces between “blocks”, major functionalities). Specify requirements for each functionality block.
- ☐ Small and partial pre-tests can help to confirm uncertainties. E.g., If you are unsure about a technology choice.
- ☐ Now you can implement the design, physically.
- ☐ The implementation is often accompanied by tests to confirm block functionality.
- ☐ Then verify that the full system works according to your goals and requirements.
- ☐ Evaluate the solution (ideas for improvements, shortcomings, ... conclusion). Make sure that you relate your conclusion to your project specification and research hypothesis, where relevant.

Evaluation criteria

Evaluation criteria for project report:

- **Product described**
 - ✓ Purpose of the system (problem/solution)
 - ✓ Instructions for use.
- **Hardware design**
 - ✓ A description of the microcontroller
 - ✓ Clear list of all the used components
 - ✓ A circuit diagram (Fritzing, draw.io)
- **Communication technologies**
 - ✓ Description of used communication technologies, e.g. wifi (theoretical background)
 - ✓ Pros and cons
- **Software design**
 - ✓ Description of the software architecture.
 - ✓ Link to own code and third party libraries that were used.
 - ✓ All code provided, incl. linkage code to IoT platforms, app design, etc.
- **Images**
 - ✓ Picture of the complete system in its real context
 - ✓ Drawing of own figures explaining crucial concepts
 - ✓ Figures and text clearly linked
- **Testing and results**
 - ✓ What kind of testing has been executed, single or multiple tests.
 - ✓ Describe the remaining weaknesses of the system.
 - ✓ Future development ideas.
- **Reflection**
 - ✓ What were the problems the team encountered?
 - ✓ How did the team overcome these problems?
 - ✓ What did the team learn while doing the project?
 - ✓ Who did what in the team, and. logbook